

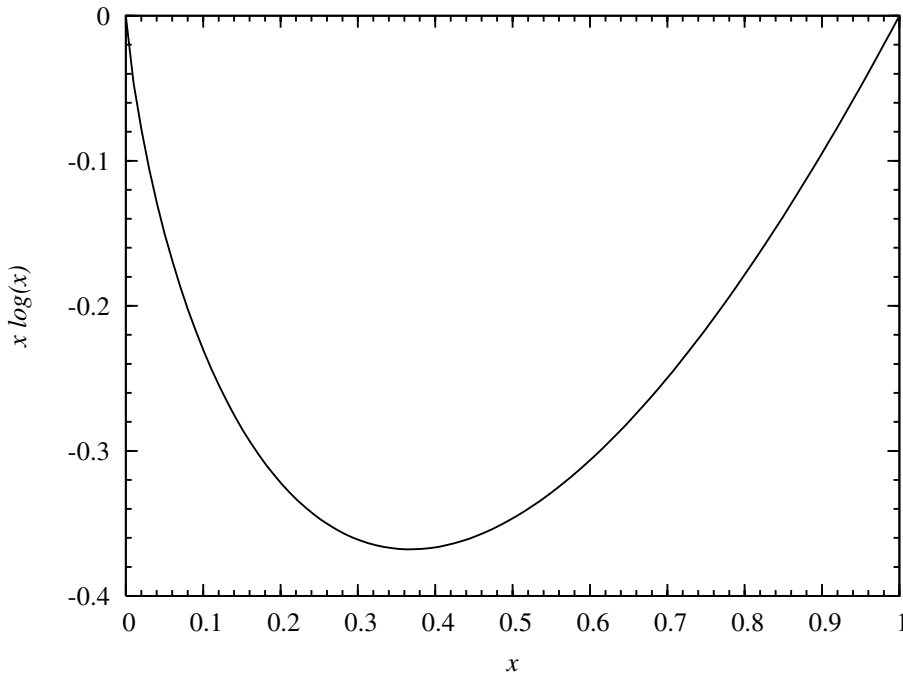


Figure 6.11.1. The function $x \log(x)$ is shown for $0 < x < 1$. Although nearly invisible, an essential singularity at $x = 0$ makes this function tricky to invert.

6.11 Inverse of the Function $x \log(x)$

The function

$$y(x) = x \log(x) \quad (6.11.1)$$

and its inverse function $x(y)$ occur in a number of statistical and information theoretical contexts. Obviously $y(x)$ is nonsingular for all positive x , and easy to evaluate. For x between 0 and 1, it is negative, with a single minimum at $(x, y) = (e^{-1}, -e^{-1})$. The function has the value 0 at $x = 1$, and it has the value 0 as its limit at $x = 0$, since the linear factor x easily dominates the singular logarithm.

Computing the inverse function $x(y)$ is, however, not so easy. (We will need this inverse in §6.14.12.) From the appearance of Figure 6.11.1, it might seem easy to invert the function on its left branch, that is, return a value x between 0 and e^{-1} for every value y between 0 and $-e^{-1}$. However, the lurking logarithmic singularity at $x = 0$ causes difficulties for many methods that you might try.

Polynomial fits work well over any range of y that is less than a decade or so (e.g., from 0.01 to 0.1), but fail badly if you demand high fractional precision extending all the way to $y = 0$.

What about Newton's method? We write

$$\begin{aligned} f(x) &\equiv x \log(x) - y \\ f'(x) &= 1 + \log(x) \end{aligned} \quad (6.11.2)$$

giving the iteration

$$x_{i+1} = x_i - \frac{x_i \log(x_i) - y}{1 + \log(x_i)} \quad (6.11.3)$$

This doesn't work. The problem is not with its rate of convergence, which is of course quadratic for any finite y if we start close enough to the solution (see §9.4). The problem is that the region in which it converges at all is very small, especially as $y \rightarrow 0$. So, if we don't already have a good approximation as we approach the singularity, we are sunk.

If we change variables, we can get different (not computationally equivalent) versions of Newton's method. For example, let

$$u \equiv \log(x), \quad x = e^u \quad (6.11.4)$$

Newton's method in u looks like this:

$$\begin{aligned} f(u) &= ue^u - y \\ f'(u) &= (1 + u)e^u \\ u_{i+1} &= u_i - \frac{u_i - e^{-u_i} y}{1 + u_i} \end{aligned} \quad (6.11.5)$$

But it turns out that iteration (6.11.5) is no better than (6.11.3).

The observation that leads to a good solution is that, since its log term varies only slowly, $y = x \log(x)$ is only very modestly curved *when it is plotted in log-log coordinates*. (Actually it is the negative of y that is plotted, since log-log coordinates require positive quantities.) Algebraically, we rewrite equation (6.11.1) as

$$(-y) = (-u)e^u \quad (6.11.6)$$

(with u as defined above) and take logarithms, giving

$$\log(-y) = u + \log(-u) \quad (6.11.7)$$

This leads to the Newton formulas,

$$\begin{aligned} f(u) &= u + \log(-u) - \log(-y) \\ f'(u) &= \frac{u + 1}{u} \\ u_{i+1} &= u_i + \frac{u_i}{u_i + 1} \left[\log\left(\frac{y}{u_i}\right) - u_i \right] \end{aligned} \quad (6.11.8)$$

It turns out that the iteration (6.11.8) converges quadratically over quite a broad region of initial guesses. For $-0.2 < y < 0$, you can just choose -10 (for example) as a fixed initial guess. When $-0.2 < y < -e^{-1}$, one can use the Taylor series expansion around $x = e^{-1}$,

$$y(x - e^{-1}) = -e^{-1} + \frac{1}{2}e(x - e^{-1})^2 + \dots \quad (6.11.9)$$

which yields

$$x \approx e^{-1} - \sqrt{2e^{-1}(y + e^{-1})} \quad (6.11.10)$$

With these initial guesses, (6.11.8) never takes more than six iterations to converge to double precision accuracy, and there is just one log and a few arithmetic operations per iteration. The implementation looks like this: